

6LOWPAN

```

No.      Time      Source                Destination            Protocol Length Info
 8451 11070.499187 da:c0:f1:9c:c9:56:3c:1e e6:db:6b:b8:8f:ac:04:7d 6LoWPAN 122 Data, Dst: e6:db:6b:b8:8f:ac:04:7d, Src: da:c0:f1:9c:c9:56:3c:1e
Frame 8451: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface 0
  Interface id: 0 (-)
    Interface name: -
    Encapsulation type: IEEE 802.15.4 Wireless PAN with FCS not present (127)
    Arrival Time: Apr 23, 2024 16:45:59.669435000 BST
    [Time shift for this packet: 0.000000000 seconds]
    Epoch Time: 1713887159.669435000 seconds
    [Time delta from previous captured frame: 17.781130000 seconds]
    [Time delta from previous displayed frame: 17.781130000 seconds]
    [Time since reference or first frame: 11070.499187000 seconds]
    Frame Number: 8451
    Frame Length: 122 bytes (976 bits)
    Capture Length: 122 bytes (976 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: wpan:6lowpan:data]
IEEE 802.15.4 Data, Dst: e6:db:6b:b8:8f:ac:04:7d, Src: da:c0:f1:9c:c9:56:3c:1e
  Frame Control Field: 0xdc71, Frame Type: Data, Frame Pending, Acknowledge Request, PAN ID Compression, Destination Addressing Mode: Long/64-bit, Frame Version: IEEE Std 802.15.4-2006, Source Addressing Mode: Long/64-bit
    ....001 = Frame Type: Data (0x1)
    ....0... = Security Enabled: False
    ....1... = Frame Pending: True
    ....1... = Acknowledge Request: True
    ....1... = PAN ID Compression: True
    ....0... = Sequence Number Suppression: False
    ....0... = Information Elements Present: False
    ....11... = Destination Addressing Mode: Long/64-bit (0x3)
    ..01... = Frame Version: IEEE Std 802.15.4-2006 (1)
    11... = Source Addressing Mode: Long/64-bit (0x3)
  Sequence Number: 15
  Destination PAN: 0x0023
  Destination: e6:db:6b:b8:8f:ac:04:7d (e6:db:6b:b8:8f:ac:04:7d)
  Extended Source: da:c0:f1:9c:c9:56:3c:1e (da:c0:f1:9c:c9:56:3c:1e)
6LoWPAN
  Fragmentation Header
    1100 0... = Pattern: First fragment (0x18)
    Datagram size: 176
    Datagram tag: 0x0007
  IPHC Header
    011... = Pattern: IP header compression (0x03)
    ...1 1... = Traffic class and flow label: Version, traffic class, and flow label compressed (0x3)
    ....1... = Next header: Compressed
    ....10... = Hop limit: 64 (0x2)
    ....0... = Context identifier extension: False
    ....1... = Source address compression: Stateful
    ....11... = Source address mode: Compressed (0x0003)
    ....0... = Multicast address compression: False
    ....0... = Destination address compression: Stateless
    ....00... = Destination address mode: Inline (0x0000)
    [Source context: 2001:db8::]
    [Origin: 489]
  Source: 2001:db8::d8c0:f19c:c956:3c1e
  Destination: 2001:630:d0:1000::d6f9
  UDP header compression
    1111 0... = Pattern: UDP compression header (0x1e)
    ....0... = Checksum: Inline
    ....00... = Ports: Inline (0)
  Source port: 5683
  Destination port: 5683
  UDP checksum: 0x554d
  Reassembled in: 8453
Data (120 bytes)
  Data: 600000000088114020010db800000000d8c0f19cc9563c1e...
  [Length: 120]
Frame (122 bytes):
  0000 71 dc 0f 23 00 7d 04 ac 8f b8 6b db e6 1e 3c 56 q..#...k...<V
  0010 c9 9c f1 c0 da c0 b0 00 07 7e 70 20 01 06 30 00 .....-p...0.
  0020 00 10 00 00 00 00 00 00 00 00 d6 f9 f0 16 33 16 33 .....3.3
  0030 55 4d 42 03 c2 46 c1 17 b4 64 61 74 61 10 ff 2b UMB..F...data..+
  0040 31 31 2e 31 2c 31 31 31 31 31 31 31 31 31 31 31 11.11,1111111111
  0050 2c 2b 32 32 2e 32 32 2c 32 32 32 32 32 32 32 32 ,+22.22,2222222222
  0060 32 32 2c 2b 33 33 2e 33 33 2c 33 33 33 33 33 33 22,+33.33,33333333
  0070 33 33 33 33 2c 2b 34 34 2e 34 3333,+44.4
Decompressed 6LoWPAN IPHC (120 bytes):
  0000 60 00 00 00 00 88 11 40 20 01 0d b8 00 00 00 00 .....@.....
  0010 d8 c0 f1 9c c9 56 3c 1e 20 01 06 30 00 d0 10 00 .....V<...0....
  0020 00 00 00 00 00 00 d6 f9 16 33 16 33 00 88 55 4d .....3.3..UM
  0030 42 03 c2 46 c1 17 b4 64 61 74 61 10 ff 2b 31 31 B..F...data..+11
  0040 2e 31 31 2c 31 31 31 31 31 31 31 31 31 31 31 31 .11,1111111111,+
  0050 32 32 2e 32 32 2c 32 32 32 32 32 32 32 32 32 32 22.22,2222222222
  0060 2c 2b 33 33 2e 33 33 2c 33 33 33 33 33 33 33 33 ,+33.33,33333333
  0070 33 33 2c 2b 34 34 2e 34 33,+44.4

```

压缩的UDP报头和收发方的CoAP Port的值(两次5683)

压缩前的UDP报头

Frame Control Field

Sequence Number & PAN ID

IPHC Header 0111 110 011 000

目的地址

UDP reassembly

Fragment 1

Datagram Size

Datagram Tag


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No.      Time      Source      Destination      Protocol Length Info
 8453 11070.544647 2001:db8::d8c0:f19c:c956:3c1e 2001:630:d0:1000::d6f9 CoAP      82      CON, MID:49734, PUT, TKN:c1 17, /
data (text/plain)
Frame 8453: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface 0
  Interface id: 0 (-)
    Interface name: -
    Encapsulation type: IEEE 802.15.4 Wireless PAN with FCS not present (127)
    Arrival Time: Apr 23, 2024 16:45:59.714895000 BST
    [Time shift for this packet: 0.000000000 seconds]
    Epoch Time: 1713887159.714895000 seconds
    [Time delta from previous captured frame: 0.007192000 seconds]
    [Time delta from previous displayed frame: 0.007192000 seconds]
    [Time since reference or first frame: 11070.544647000 seconds]
    Frame Number: 8453
    Frame Length: 82 bytes (656 bits)
    Capture Length: 82 bytes (656 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: wpan:6lowpan:ipv6:udp:coap:data-text-lines]
    [Coloring Rule Name: UDP]
    [Coloring Rule String: udp]
IEEE 802.15.4 Data, Dst: e6:db:6b:b8:8f:ac:04:7d, Src: da:c0:f1:9c:c9:56:3c:1e
  Frame Control Field: 0xdc61, Frame Type: Data, Acknowledge Request, PAN ID Compression, Destination Addressing Mode: Long.
  64-bit, Frame Version: IEEE Std 802.15.4-2006, Source Addressing Mode: Long/64-bit
    .... .001 = Frame Type: Data (0x1)
    .... .0... = Security Enabled: False
    .... .0... = Frame Pending: False
    .... .1... = Acknowledge Request: True
    .... .1.. = PAN ID Compression: True
    .... .0... = Sequence Number Suppression: False
    .... .0... = Information Elements Present: False
    .... 11... = Destination Addressing Mode: Long/64-bit (0x3)
    .... .01... = Frame Version: IEEE Std 802.15.4-2006 (1)
    .... 11... = Source Addressing Mode: Long/64-bit (0x3)
  Sequence Number: 16
  Destination PAN: 0x0023
  Destination: e6:db:6b:b8:8f:ac:04:7d (e6:db:6b:b8:8f:ac:04:7d)
  Extended Source: da:c0:f1:9c:c9:56:3c:1e (da:c0:f1:9c:c9:56:3c:1e)
6LOWPAN
  Fragmentation Header
    1110 0... = Pattern: Fragment (0x1c)
    Datagram size: 176
    Datagram tag: 0x0007
    Datagram offset: 120
    [2 Message fragments (176 bytes): #8451(120), #8453(56)]
    [Frame: 8451, payload: 0-119 (120 bytes)]
    [Frame: 8453, payload: 120-175 (56 bytes)]
    [Message fragment count: 2]
    [Reassembled 6LOWPAN length: 176]
Internet Protocol Version 6, Src: 2001:db8::d8c0:f19c:c956:3c1e, Dst: 2001:630:d0:1000::d6f9
  0110 .... = Version: 6
  .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 0000 00... = Differentiated Services Codepoint: Default (0)
  .... .... 00... = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  .... 0000 0000 0000 0000 0000 = Flow Label: 0x000000
  Payload Length: 136
  Next Header: UDP (17)
  Hop Limit: 64
  Source: 2001:db8::d8c0:f19c:c956:3c1e
  Destination: 2001:630:d0:1000::d6f9
User Datagram Protocol, Src Port: 5683, Dst Port: 5683
  Source Port: 5683
  Destination Port: 5683
  Length: 136
  Checksum: 0x554d [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
Constrained Application Protocol, Confirmable, PUT, MID:49734
  01... = Version: 1
  ..00 .... = Type: Confirmable (0)
  .... 0010 = Token Length: 2
  Code: PUT (3)
  Message ID: 49734
  Token: c117
  Opt Name: #1: Uri-Path: data
    Opt Desc: Type 11, Critical, Unsafe
    1011 .... = Opt Delta: 11
    .... 0100 = Opt Length: 4
    Uri-Path: data
  Opt Name: #2: Content-Format: text/plain; charset=utf-8
    Opt Desc: Type 12, Elective, Safe
    0001 .... = Opt Delta: 1
    .... 0000 = Opt Length: 0
    Content-type: text/plain; charset=utf-8
  End of options marker: 255
  [Response In: 8455]
  [Uri-Path: /data]
  Payload: Payload Content-Format: text/plain; charset=utf-8, Length: 115
```

Fragment 2.

CoAP-2

OF: offset 120 表明数据在复用的位置

[Payload Length: 115]

Line-based text data: text/plain (1 lines)
+11.11,1111111111,+22.22,2222222222,+33.33,3333333333,+44.44,4444444444,+55.55,5555555555,+66.66,6666666666,+77.777

Frame (82 bytes):
0000 61 dc 10 23 00 7d 04 ac 8f b8 6b db e6 1e 3c 56 a..#..}....k...<V
0010 c9 9c f1 c0 da e0 b0 00 07 0f 34 2c 34 34 34 344,4444
0020 34 34 34 34 34 34 2c 2b 35 35 2e 35 35 2c 35 35 444444,+55.55,55
0030 35 35 35 35 35 35 2c 2b 36 36 2e 36 36 2c 55555555,+66.66,
0040 36 36 36 36 36 36 36 36 36 2c 2b 37 37 2e 37 6666666666,+77.7
0050 37 37 77

Frame Control field

Sequence Number & PAN ID

Reassembled 6LoWPAN (176 bytes):
0000 60 00 00 00 00 88 11 40 20 01 0d b8 00 00 00 00@
0010 d8 c0 f1 9c c9 56 3c 1e 20 01 06 30 00 d0 10 00V<...0....
0020 00 00 00 00 00 00 d6 f9 16 33 16 33 00 88 55 4d3.3..UM
0030 42 03 c2 46 c1 17 b4 64 61 74 61 10 ff 2b 31 31 B..F...data..+11
0040 2e 31 31 2c 31 31 31 31 31 31 31 31 2c 2b 11,1111111111,+
0050 32 32 2e 32 32 2c 32 32 32 32 32 32 32 32 22.22,2222222222
0060 2c 2b 33 33 2e 33 33 2c 33 33 33 33 33 33 33 33 ,+33.33,33333333
0070 33 33 2c 2b 34 34 2e 34 34 2c 34 34 34 34 34 34 33,+44.44,444444
0080 34 34 34 34 2c 2b 35 35 2e 35 35 2c 35 35 35 35 4444,+55.55,5555
0090 35 35 35 35 35 35 2c 2b 36 36 2e 36 36 2c 36 36 555555,+66.66,66
00a0 36 36 36 36 36 36 2c 2b 37 37 2e 37 37 37 66666666,+77.777